The Box Principle — problem 4.54

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Let a_n be the number of terms in the sequence $2^1, 2^2, \ldots, 2^n$, which begin with digit 1. Prove that

$$\log 2 - \frac{1}{n} < \frac{a_n}{n} < \log 2$$

and, hence,

$$p_1 = \lim_{n\to\infty} \frac{a_n}{n} = \log 2 \approx 0.30103.$$

One sometimes says that a *randomly chosen* power of two begins with 1 with probability $\log 2 \approx 0.30103$.